Defining Effective Rosseland Mean for Expansion Opacity

Roy G. Yaron[a], Ehud Nakar[a,*]

[a] *School of Physics and Astronomy, Tel Aviv University, Tel Aviv, 6997801, Israel*

Abstract

The standard Rosseland mean does not account for the Doppler shift of photons through spectral lines during expansion and can substantially misestimate the opacity of line-rich ejecta. We define an effective Rosseland mean opacity, $\kappa_R^{\rm exp}$, for radiative transfer in expanding media where bound-bound transitions dominate the opacity. For each frequency, the method calculates the distance an outward travelling photon covers before reaching optical depth unity, and assigns an effective expansion opacity $\kappa_{\rm exp} = 1/(\rho\Delta x)$. Single-group Monte Carlo radiative transfer simulations using this opacity reproduce the trends of full frequency-group calculations for Type Ia supernova and kilonova ejecta, with peak luminosities differing by about 25%, while reducing the computational time by more than an order of magnitude.

Keywords:
opacity, Rosseland mean, expansion opacity, kilonova, supernovae

1. Introduction

The interaction between radiation and matter is primarily determined by the material properties such as composition, density, and temperature, as well as the spectral characteristics of the radiation. In expanding plasmas, due to the Doppler effect, as photons travel through the plasma, their frequency changes. This effect is mainly important for bound-bound transitions, because their narrow line structure in frequency causes photons to interact with different lines as they travel through the plasma. This effect is known as expansion opacity, and it is important for many astrophysical applications, such as Type Ia supernovae, kilonovae and stellar winds.

For non-moving plasmas, the radiation flux can be approximated well using the Rosseland mean opacity, which is a weighted average of the opacity over all frequencies. However, for expanding plasmas with significant bound-bound opacity, the Rosseland mean opacity is not a good approximation, since the width of the bb lines is negligible and photons interact with them only due to their frequency change as they travel through the plasma. Therefore the average distance a photon travels cannot be derived from $\kappa(\nu)$ alone because it inherently depends on the photon's direction of movement and the plasma's velocity gradient. To avoid this problem and to allow calculation of transport that can inherently depend on the photon's direction of movement and velocity gradient, radiation transport in systems with expansion opacity is usually calculated using Monte Carlo simulations, which track individual photons and their frequency evolution as they propagate through the expanding medium.

Calculating the Rosseland mean opacity has several motivations: a single opacity value (1) is required for analytic models and closed-form solutions, which cannot treat full frequency-dependent opacity; (2) gives direct physical intuition for how temperature and composition affect the opacity and hence the light curve; and (3) enables faster and simpler simulations than full frequency-dependent calculations.

Previous attempts to address this problem have been proposed. Fontes et al. [1, 2] suggested using artificial line broadening by assigning a line width corresponding to unphysical high temperature of 20 MeV, which effectively smears the bound-bound lines over a finite frequency range. This approach preserves the integral of the opacity over frequency (area-preserving line profiles) and for NSM ejecta it produces frequency-dependent opacities that are one to two orders of magnitude greater than those obtained with expansion opacities in the Sobolev approximation without broadening. Fontes et al. [3] demonstrated that their smeared opacity tables, calculated using the Sobolev formalism, yield light curves for multigroup radiative transfer simulations that are similar to those obtained from full line-by-line Monte Carlo radiation transport (using the superNu code) for kilonovae (neutron star mergers). However, this approach still requires multigroup frequency-dependent opacity tables rather than a single gray (frequency-averaged) Rosseland mean opacity and uses an unphysical assumption that has been verified only for specific ejecta. Several works [4, 5, 6] have shown how opacity depends on composition, temperature, density, and time by comparing the Planck mean expansion opacity over those parameters. The Planck mean is nevertheless not the appropriate average for the opacity that controls photon escape from the ejecta. In a non-expanding medium this transport opacity is the Rosseland mean, which weights the low-opacity frequency windows through which photons preferentially diffuse. In an expanding medium the same transport character is needed, but the opacity must also include the line absorption encountered as the photon redshifts

*Corresponding author: udini@tauex.tau.ac.il

through the expanding flow.

In this paper we present the effective expansion opacity, $\kappa_{\rm exp}$, which accounts for bound-bound expansion opacity. The method follows an outward-moving photon in homologous expansion ($r = vt$) and calculates the distance over which it reaches optical depth $\tau \simeq 1$. Taking the Rosseland mean of $\kappa_{\rm exp}$ defines the effective expansion Rosseland mean, $\kappa_R^{\rm exp}(T, \rho, t)$. This gray-opacity method assumes that the photons dominating the diffusion are close to a blackbody spectrum. It is therefore useful for bolometric luminosities and characteristic temperatures, but not for detailed line spectra.

The structure of the paper is as follows: In Section 2 we describe how the expansion opacity is calculated, and define the effective expansion opacity. In Section 3, we show the variation of $\kappa_R^{\rm exp}$ with composition, temperature, density, and time and show how it scales differently from the Planck mean. In Section 4 we demonstrate the use of $\kappa_R^{\rm exp}$ in radiative transfer simulations and compare the results to the full frequency-group calculations. In Section 5 we summarize the results.

2. Expansion Opacity

The total opacity κ is the sum of contributions from different physical processes:

$$\kappa = \kappa_{\rm T} + \kappa_{\rm ff} + \kappa_{\rm bf} + \kappa_{\rm bb} \tag{1}$$

where $\kappa_{\rm T}$ is Thomson scattering, $\kappa_{\rm ff}$ is free-free opacity, $\kappa_{\rm bf}$ is bound-free opacity (which we neglect in this work), and $\kappa_{\rm bb}$ is bound-bound opacity from atomic line transitions.

The bb opacity $\kappa_{\rm bb}$ depends on the ionization and excitation populations of the material. We obtain the ionization fractions from the Saha equation and the level populations from the Boltzmann distribution, and use them to calculate the optical depth of each line.

The optical depth of the bound-bound line is then calculated using the Sobolev approximation [7].

$$\tau_{j_l,j_u} = \frac{\pi e^2}{m_e} \frac{f_{j_l,j_u}}{\nu_{j_l,j_u}} t \left(N_{m,i,j_l} - \frac{g_{j_l}}{g_{j_u}} N_{m,i,j_u} \right) \tag{2}$$

where m is the material index, i is the ionization state, j_l and j_u are the lower and upper energy levels respectively, f_{j_l,j_u} is the oscillator strength of the transition from the energy level j_l to the energy level j_u, ν_{j_l,j_u} is the frequency of the transition, N_{m,i,j_l} and N_{m,i,j_u} are the number densities of the lower and upper levels respectively, g_{j_l} and g_{j_u} are their degeneracies, and t is the time from explosion.

There are two values for the bound-bound opacity, saturated and unsaturated which are calculated as:

$$\kappa_{bb,saturated}(\nu) = \sum_l \frac{\nu_l}{\rho ct} \delta(\nu - \nu_l) \cdot (1 - \exp(-\tau_{j_l,j_u})) \tag{3}$$

$$\kappa_{bb,unsaturated}(\nu) = \sum_l \frac{\nu_l}{\rho ct} \delta(\nu - \nu_l) \cdot \tau_{j_l,j_u} \tag{4}$$

The saturated opacity gives the contribution of a narrow line to a larger finite frequency bin, by capping its effect once it becomes optically thick, since a single line has a maximal effect on the opacity once all photons in the line's frequency are absorbed. The unsaturated opacity gives the optical depth accumulated by a photon that crosses the full frequency bin.

We define the effective expansion opacity, $\kappa_{\rm exp}$, from the distance an outward travelling photon moves through the plasma before accumulating optical depth unity. The effective expansion Rosseland mean, $\kappa_R^{\rm exp}$, is then the Rosseland mean of $\kappa_{\rm exp}$. An outward travelling photon with frequency ν_0, will reach a new spectral line with frequency ν_l after a velocity shift of $\Delta v = -c \ln(\nu_0/\nu_l)$ which corresponds to a distance of $\Delta x = -ct \ln(\nu_0/\nu_l)$.

For that we define the cumulative frequency optical depth as:

$$\tau(\nu) = \int \kappa(\nu) \rho \, dx = ct \int \kappa(\nu) \rho \frac{d\nu}{\nu} \tag{5}$$

And for each frequency ν_i, given the frequency ν_j so that $\tau(\nu_j) - \tau(\nu_i) = 1$, we define the effective expansion opacity as:

$$\kappa_{\rm exp}(\nu_i) = \frac{1}{\rho \Delta x} = \frac{1}{\rho ct \ln(\nu_j/\nu_i)} \tag{6}$$

For finite size frequency bins this process is done by estimating where inside the stopping bin the photon reaches optical depth unity. This is needed because assigning the stopping point to the edge of a frequency bin would make the minimum travel distance depend on the chosen bin width, rather than on the line opacity inside the bin.

Numerically, after discretizing the frequency, the stopping bin is found with the unsaturated optical depth by finding the first bin j such that

$$\tau(\nu_{j-1}) - \tau(\nu_i) < 1, \qquad \tau(\nu_j) - \tau(\nu_i) > 1 \tag{7}$$

Unsaturated opacity is used for this step because it identifies where the photon reaches true optical depth unity; with saturated opacity alone, a strong line with $\tau \gg 1$ contributes only $1 - \exp(-\tau)$ and cannot set the stopping point by itself. We then estimate the sub-bin stopping frequency in bin j using the saturated opacity. If the saturated optical depth in this bin can supply the remaining optical depth to unity, we use

$$\begin{aligned} \Delta\nu_{\tau=1,i} &= \nu_{j-1} - \nu_i + (\nu_j - \nu_{j-1}) \frac{1 - (\tau(\nu_{j-1}) - \tau(\nu_i))}{\Delta\tau_j^s}, \\ \nu_{\tau=1,i} &= \nu_i + \Delta\nu_{\tau=1,i}. \end{aligned} \tag{8}$$

where $\Delta\tau_j^s$ is the saturated optical depth of bin j. Otherwise, if $\Delta\tau_j^s < 1 - (\tau(\nu_{j-1}) - \tau(\nu_i))$, we set $\Delta\nu_{\tau=1,i} = \nu_j - \nu_i$. The effective opacity is then calculated from $\Delta x_i = ct \ln(\nu_{\tau=1,i}/\nu_i)$. When a single bin dominates, we instead use

$$\kappa_{\rm exp}(\nu_i) = \frac{\sum_l (1 - \exp(-\tau(\nu_l)))}{\rho ct \ln(\nu_{i+1}/\nu_i)} + \kappa_{\rm T} + \kappa_{ff}(\nu_i) \tag{9}$$

This interpolation is approximate, but it is accurate when $\Delta\tau \ll 1$ or $\Delta\tau \gg 1$. For the intermediate case, $\Delta\tau \sim 1$, it introduces a small error which should mostly cancel on average.

Figure 1 shows the unsaturated opacity, saturated opacity, and effective expansion opacity of iron after 3 days, for density 1×10^{-12} g/cm^3 and temperature 0.5 eV. For species with $A < 28$ in Type Ia supernova compositions we use the atomic line data of Kurucz & Bell (1995; CD 23) [8]. For kilonova composition, where the relevant nuclei have $A > 28$, we use the NIST Atomic Spectra Database [9].

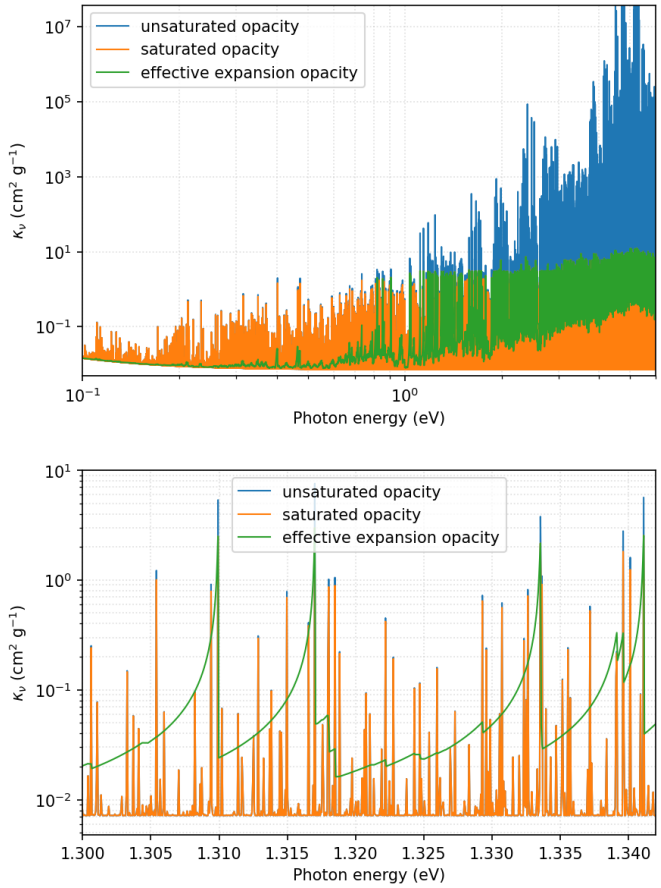


Figure 1: Unsaturated opacity, saturated opacity, and effective expansion opacity, for iron at $\rho = 1 \times 10^{-12}$ g/cm^3, $T = 0.5$ eV, $t = 3$ days. For large (top) and zoomed (bottom) energy ranges.

3. Rosseland Mean of Effective Expansion Opacity

The Planck mean opacity is an emission-weighted average, while the Rosseland mean opacity is a transport-weighted harmonic mean:

$$\kappa_P = \frac{\int \kappa_\nu B_\nu(T) d\nu}{\int B_\nu(T) d\nu}, \qquad \frac{1}{\kappa_R} = \frac{\int \frac{1}{\kappa_\nu} \frac{\partial B_\nu(T)}{\partial T} d\nu}{\int \frac{\partial B_\nu(T)}{\partial T} d\nu}. \tag{10}$$

where $B_\nu(T)$ is the Planck function. When the Rosseland average is applied to the usual frequency-dependent opacity κ_ν we denote the result by κ_R. When it is applied to the effective expansion opacity $\kappa_{\rm exp}$, we denote the result by $\kappa_R^{\rm exp}$.

The main problem with applying the Rosseland mean directly to the expansion opacity is that zero-width bound-bound lines cannot contribute to a frequency-integrated harmonic mean; numerically, this appears as a vanishing bound-bound contribution as the frequency bins are refined. Figure 2 shows the convergence of the Rosseland mean opacities as a function of the frequency resolution $\Delta\nu/\nu$ (for logarithmically spaced frequency bins) for iron (Fe, $Z = 26$) in top and erbium (Er, $Z = 68$) in bottom. Iron is an iron-group element with an open d-block, representative of the heavy material in Type Ia supernova ejecta. Erbium is a lanthanide with an open f-block, which is common in kilonova compositions and gives a much larger bound-bound opacity than iron. In Figure 2 we plot the ordinary Rosseland mean of the saturated expansion opacity, κ_R, and the effective expansion Rosseland mean, $\kappa_R^{\rm exp}$.

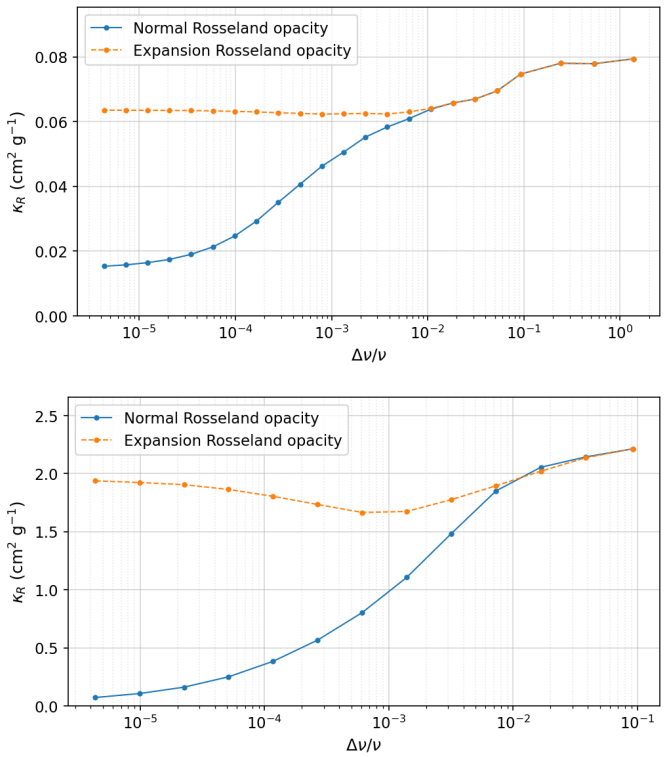


Figure 2: Convergence of the Rosseland mean opacities vs. the frequency resolution in $\Delta\nu/\nu$. Top: iron for Type Ia-like conditions, $\rho = 1 \times 10^{-13}$ g/cm^3, $T = 1$ eV, $t = 20$ days. Bottom: erbium (lanthanide, $Z = 68$); $\rho = 1 \times 10^{-12}$ g/cm^3, $T = 1$ eV, $t = 1$ day. In each panel the blue solid curve is the ordinary Rosseland mean κ_R of the saturated expansion opacity, and the orange dashed curve is the effective expansion Rosseland mean $\kappa_R^{\rm exp}$.

The results show that $\kappa_R^{\rm exp}$ is barely dependent on the frequency resolution and gives converged opacity even for low resolutions such as $\Delta\nu/\nu \gtrsim 4 \cdot 10^{-2}$ (which is equivalent to less than 200 bins for the full spectrum used). The reason the effective opacity gives correct results even for large bins is that the spectral lines are randomly distributed within each frequency bin. As long as the lines are randomly distributed, Eq. (9) provides a good approximation since it represents the typical distance a photon travels in a frequency bin. The fact that we observe good agreement between low-resolution and high-resolution calculations for $\kappa_R^{\rm exp}$ for both iron and erbium confirms that the line locations are indeed close enough to

randomly distributed within large bins. This also offers a rationale for the accuracy of the results done with the artificial line-broadening prescription of Fontes et al. [1, 2]: spreading line opacity over frequency is analogous to many narrow lines at random positions within a coarse bin, so either picture can yield similar cumulative opacity in the bin, consistent with smeared Sobolev tables reproducing Monte Carlo transport [3]. At low frequency resolution κ_R of the saturated expansion opacity and $\kappa_R^{\rm exp}$ are close because a single wide bin often sets the stopping distance. In that limit, Eq. (9) applies and the effective opacity is the same as the saturated opacity. Using the formalism formerly described, we calculated the opacity of materials for different times, densities and temperatures and constructed opacity tables that can be used for simulations of type Ia supernova and kilonova. To simulate Type Ia supernova ejecta, we created two tables for two compositions. The first for the outer layer of the supernova, which is a mixture of Silicon, Sulfur, Argon, and Calcium in the ratio of 0.535, 0.32, 0.062, and 0.083 respectively. The second for the core of the supernova, which is a decay of ^{56}Ni followed its decay chain, $^{56}\mathrm{Ni} \rightarrow {}^{56}\mathrm{Co} \rightarrow {}^{56}\mathrm{Fe}$, using the composition appropriate for each time window. We assumed the half-lives of $t_{1/2}(^{56}\mathrm{Ni}) = 6.075$ days and $t_{1/2}(^{56}\mathrm{Co}) = 77.233$ days. For the kilonova simulation we created a single table for a reduced mixture, $X_{34} = 0.5$, $X_{44} = 0.15$, $X_{52} = 0.25$, and $X_{68} = 0.1$. Kilonova ejecta contain many elements, so this mixture was chosen to represent the main p-, d-, and f-block contributions without calculating every species. The block fractions were estimated from the $Y_e = 0.2$ abundance distribution in Fig. 11 of Tanaka et al. [6]; Se ($Z = 34$) and Te ($Z = 52$) represent the p-block, Ru ($Z = 44$) the d-block, and Er ($Z = 68$) the f-block. Within each block we selected elements with relatively large abundance.

Each table stores, at every (T, ρ, t), $\kappa_R^{\rm exp}$, $f_{\rm abs}$, $\kappa_P^{\rm abs}$, and $\kappa_P^{\rm tot}$. The fraction $f_{\rm abs}$ is the Rosseland mean of $\kappa_{\rm abs}/(\kappa_{\rm abs} + \kappa_{\rm scat})$ with $\kappa_{\rm abs} = \kappa_{\rm exp} - \kappa_{\rm T}$ and $\kappa_{\rm scat} = \kappa_{\rm T}$,

$$f_{\rm abs} = \frac{\int \frac{\kappa_{\rm abs}}{\kappa_{\rm abs}+\kappa_{\rm scat}} \frac{\partial B_\nu}{\partial T}\, d\nu}{\int \frac{\partial B_\nu}{\partial T}\, d\nu}. \tag{11}$$

In transport we use $\kappa_R^{\rm abs} = f_{\rm abs}\, \kappa_R^{\rm exp}$ and $\kappa_R^{\rm scat} = (1 - f_{\rm abs})\, \kappa_R^{\rm exp}$: absorption deposits energy and re-emits the packet (IMC), while scatter only redirects it. The Planck mean $\kappa_P^{\rm abs}$ is taken from saturated absorption, $\kappa_{\rm sat} - \kappa_{\rm T}$, and sets the cell temperature from the deposited energy; $\kappa_P^{\rm tot}$ is stored only for comparison with $\kappa_R^{\rm exp}$.

Since the ordinary Rosseland mean opacity is problematic to define for expansion opacity, previous works [4, 5, 6] often used the Planck mean opacity to compare the opacity of different compositions in different temperatures. We compare $\kappa_R^{\rm exp}$ to the Planck mean through their ratio as a function of temperature and density. For consistency in this comparison we use the Planck mean of the total saturated opacity, $\kappa_P^{\rm tot}$, including Thomson scattering, so that both means contain the same physical processes and the ratio tests how much the frequency weighting differs rather than whether scattering is included in one mean but not the other. Figure 3, Figure 4, and Figure 5 show $\kappa_R^{\rm exp}$ and the ratio $\kappa_P^{\rm tot}/\kappa_R^{\rm exp}$ for the ^{56}Ni decay chain, Type Ia outer-layer composition, and kilonova composition. For the Ni-rich core and kilonova compositions the ratio varies approximately between 1 and 30 and depends on density and temperature, so $\kappa_P^{\rm tot}$ is not strongly correlated with $\kappa_R^{\rm exp}$. For the Type Ia outer-layer composition (Si, S, Ar, Ca) the ratio stays between about 1 and 2 over the plotted density range (10^{-14}–10^{-10} g/cm^3). In this regime the opacity is dominated by Thomson scattering and free-free absorption, which depend only weakly on frequency, so the two averaging schemes give similar results. Figures 6 and 7 show one-dimensional slices of $\kappa_R^{\rm exp}$ for the Type Ia outer-layer and kilonova compositions. The kilonova opacity is larger than the Type Ia opacity over most of the plotted range, especially at low density and high temperature, because of the stronger bound-bound contribution from the heavier elements. For both compositions the opacity generally decreases with density, while the temperature dependence is weaker for the Type Ia composition and stronger for the kilonova composition.

4. Simulations With Single Group Opacity Tables

We performed radiation transport simulations to test the effective expansion Rosseland opacity model. The simulations were performed using URILIGHT [10, 11, 12], a Monte Carlo radiative transfer code written by Elbaz et al. and designed for supernova ejecta. For this work we made several modifications to the code, including support for Rosseland and Planck mean opacity tables and related changes needed for the simulations presented here. With permission from the code writers, we published the modified version in a Git repository (`https://github.com/roee2000/Urilight_SN`). The Type Ia supernova model is one-dimensional and spherically symmetric and contains 100 cells in homologous expansion. The ejecta has two composition layers: an inner $0.98\ M_\odot$ layer of ^{56}Ni and an outer $0.4\ M_\odot$ layer of Si, S, Ar, and Ca. The simulation starts at $t = 2$ days and ends at $t = 80$ days with 100 logarithmic time steps.

Energy is deposited by the $^{56}\mathrm{Ni} \rightarrow {}^{56}\mathrm{Co} \rightarrow {}^{56}\mathrm{Fe}$ decay chain. Gamma-rays and positrons are transported at the beginning of the calculation, and their deposited energy is saved for each cell and time step. In the radiation transport stage, new optical photon packets are emitted according to the fraction of the total gamma-ray and positron energy deposited in each cell. Existing photon packets are propagated with effective scattering: when a packet is absorbed it is not removed, but is re-emitted with a frequency sampled according to the local cell emission spectrum.

The kilonova simulation uses a one-dimensional ejecta profile with total mass $0.05\ M_\odot$ and 32 cells, from $t = 0.05$ days to $t = 15$ days with 55 time steps. The composition is uniform and contains Se, Ru, Te, and Er with mass fractions 0.5, 0.15, 0.25, and 0.1, respectively. Instead of a radioactive decay chain, the heating is an external power law per unit mass [13], $\dot{\epsilon} = 10^{10}(t/1\ \mathrm{day})^{-1.3}\ \mathrm{erg\ s^{-1}\ g^{-1}}$.

In the full frequency-group calculation, photon packets keep their frequency and interact with the frequency-dependent

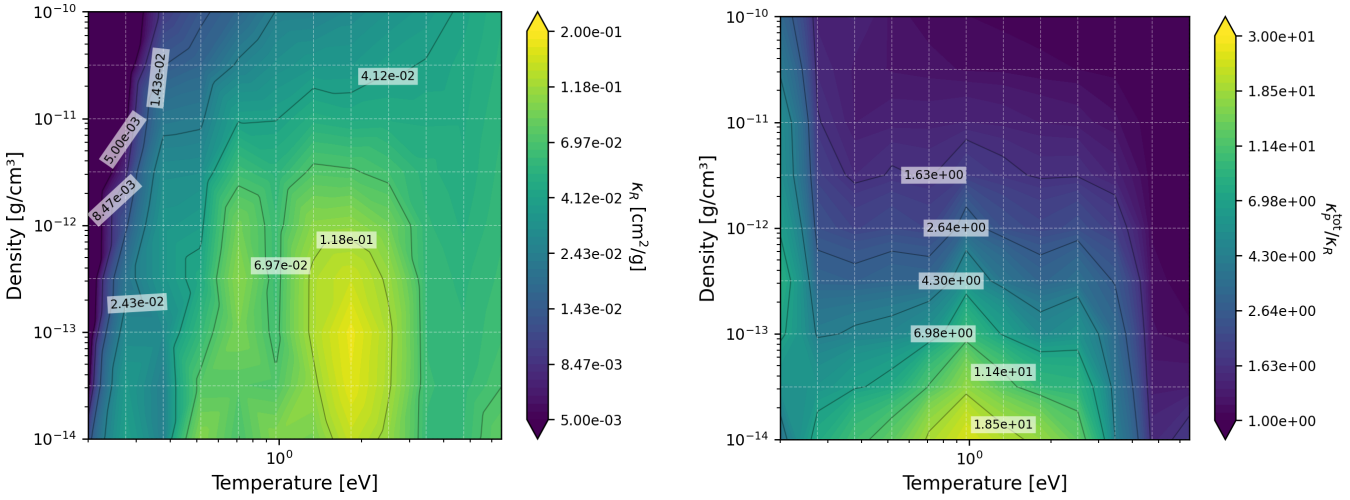


Figure 3: Type Ia core composition at $t = 10$ days: $\kappa_R^{\rm exp}$ (left) and $\kappa_P^{\rm tot}/\kappa_R^{\rm exp}$ ratio (right). Density 10^{-14}–10^{-10} g/cm^3.

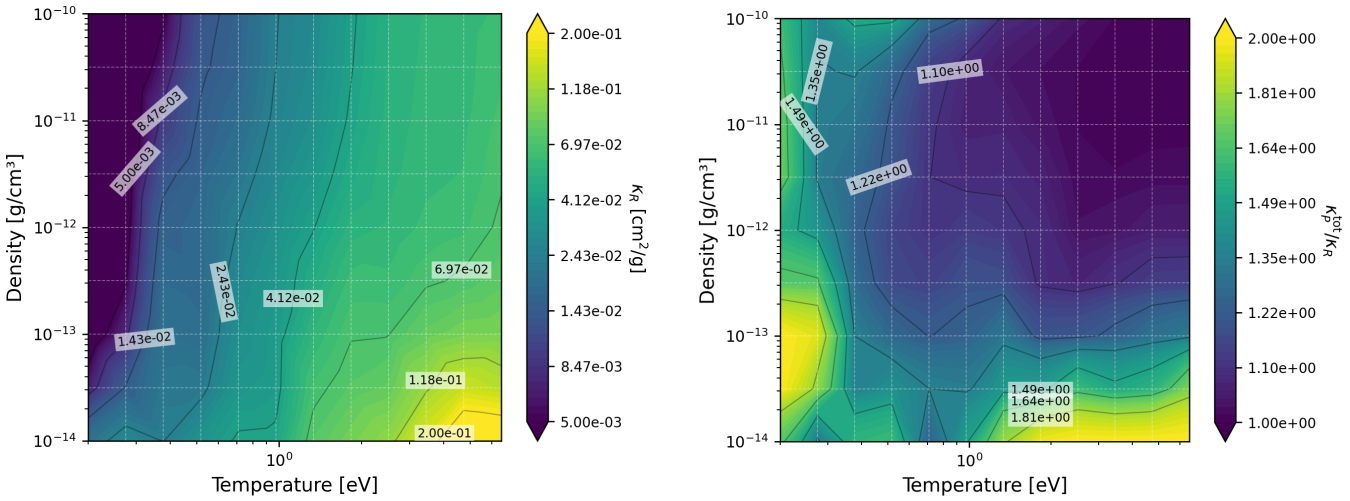


Figure 4: Type Ia outer-layer composition at $t = 10$ days: $\kappa_R^{\rm exp}$ (left) and $\kappa_P^{\rm tot}/\kappa_R^{\rm exp}$ ratio (right). The composition is Si, S, Ar, and Ca with mass fractions 0.535, 0.32, 0.062, and 0.083. Density 10^{-14}–10^{-10} g/cm^3. The ratio stays near unity because the opacity is dominated by Thomson scattering and free-free absorption.

opacity groups. For that, at each time step the simulation calculates the ionization stages of each material and goes through all bound-bound lines to calculate the opacity for that temperature and density. In the single-group calculation, the photon mean free path is set by $\kappa_R^{\rm exp}$, interpolated from the opacity table using local (T, ρ, t). The table value $f_{\rm abs}$ splits this into absorbing and scattering Rosseland means ($\kappa_R^{\rm abs}$ and $\kappa_R^{\rm scat}$) as described above. The cell temperature is determined by requiring the emitted energy to match the deposited energy; for this step we use $\kappa_P^{\rm abs}$, the Planck mean of the saturated absorption opacity.

The ejecta geometries used for the validation are shown in Figure 8. Both simulations are one-dimensional homologous expansions, so the spatial coordinate can be represented by the velocity coordinate $r = vt$. The Type Ia model uses an exponential density profile matching the W7 model [14], $\rho \propto t^{-3}\exp(-v/v_e)$, with $v_e = 2.7 \times 10^8$ cm s^{-1} ($0.009c$) and an outer edge at $0.093c$. The kilonova model uses a flat profile $\rho = \rho_0$ for $v < 0.1c$ and $\rho = \rho_0(v/0.1c)^{-4}$ for $0.1c < v < 0.4c$, with ρ_0 set so the total mass is $0.05\ M_\odot$. The v^{-4} outer slope follows the ejecta-density consideration in [15].

Figure 9 compares the light curves obtained with the single-group calculation, using $\kappa_R^{\rm exp}$, to the full frequency-group calculation for both ejecta models. The one-group and multigroup results follow the same qualitative trends in both cases, showing that the averaged opacity captures the main transport behavior while avoiding the cost of recalculating the full frequency-dependent opacity during the transport calculation.

Figure 10 maps $T_{\rm plasma}$ and $\kappa_R^{\rm exp}$ in the Type Ia validation simulation on the (v, t) grid. The temperature is taken from the URILIGHT profiles. The opacity in each cell is interpolated from the precomputed NiCoFe and SN1A tables using local (T, ρ, t), with ρ from homologous expansion and the ejecta

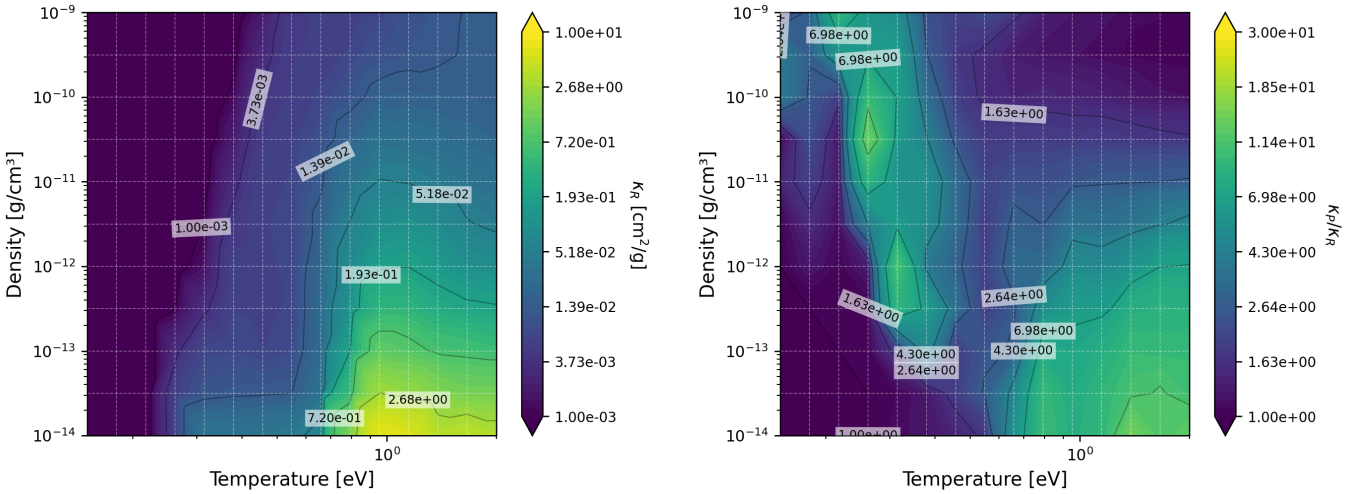


Figure 5: Kilonova composition at $t = 1$ day: $\kappa_R^{\rm exp}$ (left) and $\kappa_P^{\rm tot}/\kappa_R^{\rm exp}$ ratio (right). Density 10^{-14}–10^{-9} g/cm^3.

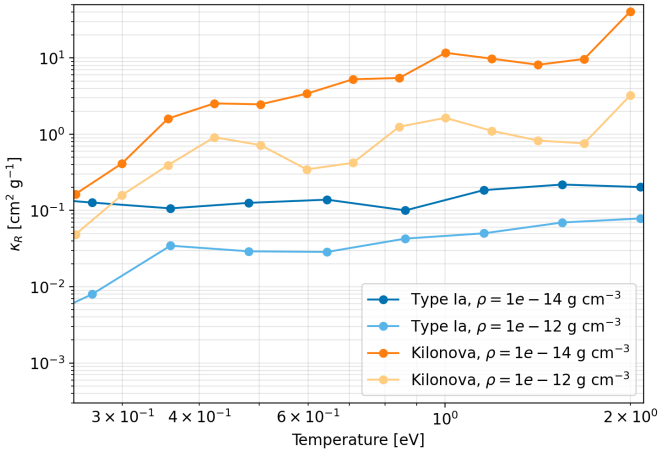


Figure 6: Effective expansion Rosseland mean, $\kappa_R^{\rm exp}$, as a function of temperature for fixed densities $\rho = 10^{-14}$ and 10^{-12} g/cm^3. The plotted table times are $t = 1$ day for Type Ia and $t = 1.35721$ days for the kilonova. Type Ia outer-layer composition is shown in blue and kilonova composition in orange. Dark and light curves correspond to the lower and higher density, respectively.

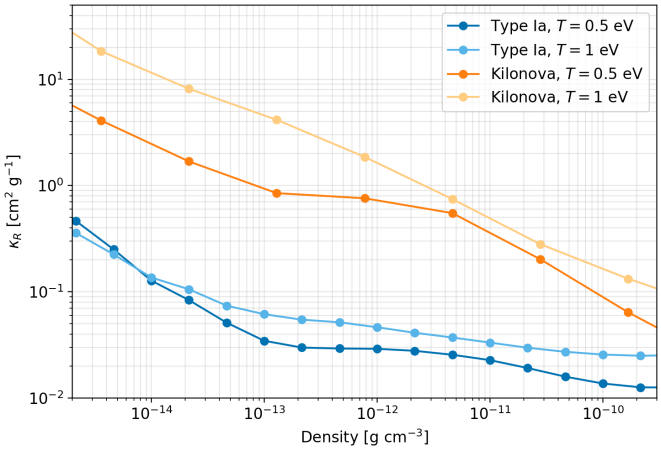


Figure 7: Effective expansion Rosseland mean, $\kappa_R^{\rm exp}$, as a function of density for fixed temperatures $T = 0.5$ and 1 eV. The plotted table times are $t = 1$ day for Type Ia and $t = 1.35721$ days for the kilonova. Type Ia outer-layer composition is shown in blue and kilonova composition in orange. Dark and light curves correspond to the lower and higher temperature, respectively.

geometry. The Ni/IME boundary is marked by a black dashed line. With time, the ejecta cools and the outer layers become more transparent.

On the opacity panel, the red curve is the $\tau = 1$ surface: at each t we integrate inward from the outer edge until $\int \kappa_R^{\rm exp}\, \rho\, dR_{\rm eff} = 1$, where each velocity shell has radial thickness $\Delta R_i = t\, \Delta v_i$. Because photons diffuse isotropically in three dimensions, we use an effective path $\Delta R_{\rm eff} = \sqrt{3}\, \Delta R$ when evaluating the opacity integral in this spherical geometry.

The cyan dashed curve marks where the random-walk photon leakage time equals the dynamical time, $t_{\rm phot} = t$. Using the same inward integration, the cumulative optical depth to shell i is $\tau_i = \tau_{i+1} + \kappa_{R,i}^{\rm exp}\, \rho_i\, \Delta R_{{\rm eff},i}$, with $\tau = 0$ at the outer edge. A diffusing photon does not cross a shell once: each scatter can send it back, so on average it traverses the thickness $(2\tau_i + 1)$ times before leaking outward. Each traversal takes time $\Delta R_i/c$, so

$$\Delta t_i = \frac{(2\tau_i + 1)\, \Delta R_i}{c}. \tag{12}$$

Starting from the outer edge and accumulating inward, we locate the velocity where $t_{\rm phot} = t$. Together, the two curves mark the last-scattering surface and the diffusion depth relevant at time t.

4.1. *Runtime Advantage of Single-Group Transport*

The single-group calculation is faster for three main reasons. First, each photon packet samples a Rosseland-mean transport opacity, so fewer packets are needed to obtain a stable estimate of the frequency-averaged energy transport. In a multigroup calculation, the same average behavior is recovered only statistically from packets distributed over frequency,

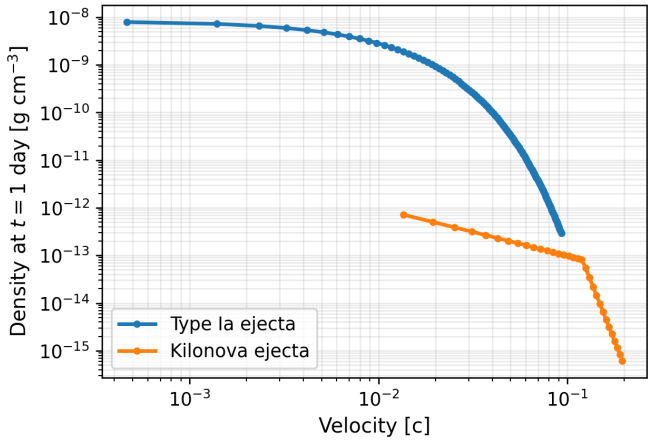


Figure 8: Density profiles of the Type Ia and kilonova ejecta at $t = 1$ day. The simulations assume spherical homologous expansion, so each shell keeps a fixed velocity and its radius grows as $r = vt$. The Type Ia model is centrally concentrated and slower, while the kilonova model occupies faster, lower-density ejecta.

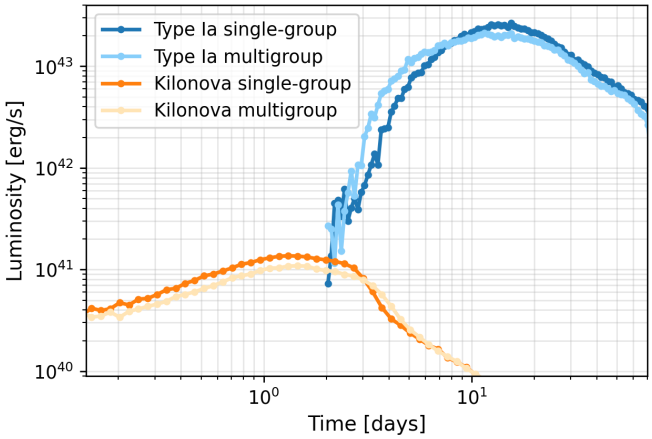


Figure 9: Bolometric luminosity from the Type Ia and kilonova validation simulations. For each ejecta model we compare the single-group calculation using $\kappa_R^{\rm exp}$ with the corresponding full frequency-group calculation.

which requires more packets to reduce numerical noise. Second, each packet propagation is cheaper. In a line-rich multigroup calculation, photons can repeatedly redshift into dense groups of bound-bound lines, be absorbed after very short path lengths, and be re-emitted until they enter a more transparent frequency window. These repeated short propagation steps add many microsteps inside a single time step. The single-group calculation replaces this behavior with an averaged mean free path, so packet histories contain fewer, more uniform transport steps. Third, the multigroup calculation must recompute the frequency-dependent opacity for every cell from its local (T, ρ, t) by going through the bound-bound line list of the material. The single-group calculation instead interpolates one value from a precomputed opacity table.

The speedup depends on which part of the multigroup calculation is the bottleneck. For lanthanide-rich kilonova ejecta, where photon transport through many bound-bound lines can dominate the runtime, the single-group calculation can be about 30 times faster and has significantly smaller numerical noise for the same number of packets. In cases where the main multigroup cost is recalculating the opacity itself, the gain can be even larger; for example, the Type Ia calculation shown in Figure 9 was about 100 times faster in the single-group calculation. This advantage is strongest when the same composition is used many times, since the opacity table can be reused. If many different compositions must be explored, a separate table has to be built for each composition, and this preprocessing can itself require significant runtime.

5. Summary

We defined the effective expansion opacity, $\kappa_{\rm exp}$, for media in homologous expansion. The opacity is defined from the average distance an outward travelling photon needs to reach optical depth unity, $\kappa_{\rm exp} = 1/(\rho\Delta x)$. Taking the Rosseland mean of $\kappa_{\rm exp}$ gives $\kappa_R^{\rm exp}$, which is not dependent on the frequency resolution. We showed this for both iron and erbium, representing iron-group Type Ia material and lanthanide-rich kilonova material. We also compared $\kappa_R^{\rm exp}$ with the Planck mean opacity for Type Ia core, Type Ia outer-layer, and kilonova compositions. The ratio between the two depends strongly on temperature and density, showing that the Planck mean is not a reliable replacement for $\kappa_R^{\rm exp}$.

Finally, we tested single-group radiation transport simulations using the new opacity tables in URILIGHT. The single-group light curves reproduce the main behavior of full frequency-group calculations for both Type Ia supernova and kilonova ejecta. The remaining differences are expected because a gray treatment cannot capture all effects of temperature gradients and frequency-dependent transport.

The Python code for building opacity tables using this algorithm is available at `https://github.com/roee2000/expansion_opacity_calculation`.

References

[1] C. J. Fontes, C. L. Fryer, A. L. Hungerford, P. Hakel, J. Colgan, D. P. Kilcrease, M. E. Sherrill, Relativistic opacities for astrophysical applications, High Energy Density Physics 16 (C) (2015) 53–59.

[2] C. J. Fontes, C. L. Fryer, A. L. Hungerford, R. T. Wollaeger, S. Rosswog, E. Berger, A line-smeared treatment of opacities for the spectra and light curves from macronovae, arXiv preprint arXiv:1702.02990 (2017).

[3] C. J. Fontes, C. L. Fryer, A. L. Hungerford, R. T. Wollaeger, O. Korobkin, A line-binned treatment of opacities for the spectra and light curves from neutron star mergers, Monthly Notices of the Royal Astronomical Society 493 (3) (2020) 4143–4171.

[4] J. Deprince, G. Wagle, S. Ben Nasr, H. Carvajal Gallego, O. Just, P. Palmeri, P. Quinet, M. Godefroid, S. Van Eck,

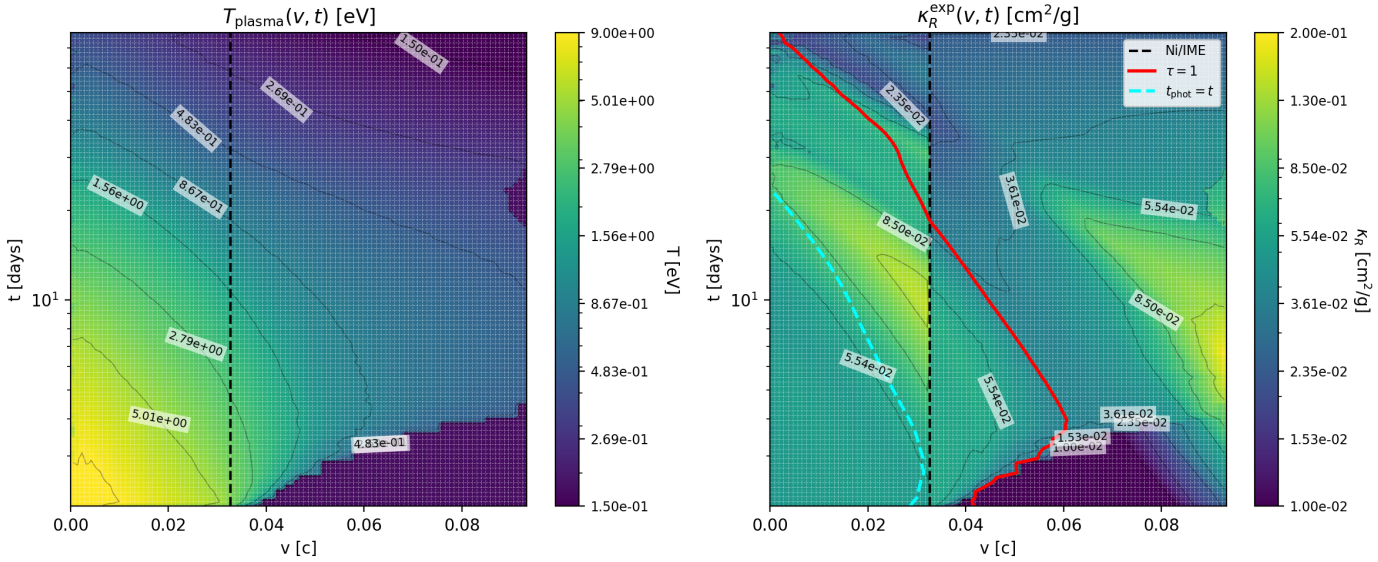


Figure 10: $T_{\rm plasma}$ (left) and $\kappa_R^{\rm exp}$ (right) in the Type Ia validation simulation. The black dashed line marks the Ni/IME boundary. On the opacity map, the red and cyan curves show the $\tau = 1$ surface and the $t_{\rm phot} = t$ surface (see text).

Kilonova ejecta opacity inferred from new large-scale hfr atomic calculations in all elements between ca (z = 20) and lr (z = 103), Astronomy & Astrophysics (2024).

[5] O. Just, et al., Properties of kilonovae from dynamical and post-merger ejecta of neutron star mergers, The Astrophysical Journal (2022).

[6] M. Tanaka, D. Kato, G. Gaigalas, K. Kawaguchi, Systematic Opacity Calculations for Kilonovae, The Astrophysical Journal 891 (2) (2020) 143.

[7] R. G. Eastman, P. A. Pinto, Spectrum formation in supernovae: Numerical techniques, The Astrophysical Journal 412 (1993) 731. doi:10.1086/172957.

[8] R. L. Kurucz, B. Bell, Atomic line list (1995).

[9] Kramida, A., Ralchenko, Yu., Reader, J., NIST ASD Team, NIST Atomic Spectra Database, National Institute of Standards and Technology, Gaithersburg, MD; version 5.11 (2024).

[10] Y. Elbaz, URILIGHT: Time-dependent Monte-Carlo radiative-transfer, Astrophysics Source Code Library, ascl:2209.012 (2022).
URL https://ascl.net/2209.012

[11] N. Wygoda, Y. Elbaz, B. Katz, Type Ia supernovae have two physical width-luminosity relations and they favour sub-Chandrasekhar and direct collision models - I. Bolometric, Monthly Notices of the Royal Astronomical Society 484 (2019). doi:10.1093/mnras/stz145.

[12] N. Wygoda, Y. Elbaz, B. Katz, Type Ia supernovae have two physical width-luminosity relations and they favour sub-Chandrasekhar and direct collision models - II. Colour evolution, Monthly Notices of the Royal Astronomical Society 484 (2019). doi:10.1093/mnras/stz146.

[13] O. Korobkin, S. Rosswog, A. Arcones, C. Winteler, On the astrophysical robustness of the neutron star merger r-process, Monthly Notices of the Royal Astronomical Society 426 (3) (2012) 1940–1949. doi:10.1111/j.1365-2966.2012.21872.x.

[14] K. Nomoto, F.-K. Thielemann, K. Yokoi, Accreting white dwarf models of type I supernovae. III. carbon deflagration supernova, The Astrophysical Journal 286 (1984) 644. doi:10.1086/162645.

[15] K. Hotokezaka, E. Nakar, Radioactive heating rate of r-process elements and macronova light curve, The Astrophysical Journal 891 (2) (2020) 152. doi:10.3847/1538-4357/ab764d.